

Toward using Digital Phenotypes in Mobile Tactile Stimulation for Children with ASD

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ABSTRACT

86% of children with Autism Spectrum Disorders (ASD) exhibit tactile impairments. Recent research has shown Mobile Tactile Stimulation is effective as a sensory processing therapy for children with ASD. The amount of time, duration, and intensity of such tactile stimulation depends on the appreciation of therapists and their interpretability based on experience and judgment. So, there is an untapped potential to propose an approach that can objectively quantify progress by leveraging the capabilities of Mobile Tactile Stimulation to gather continuous streams of data from touch interactions. In this position paper, we present a proposal to use Digital Phenotypes as an approach to adjust the “digital active ingredients” of tactile stimulation. We discuss our vision towards using Digital Phenotypes as a tool to follow up patient trajectories and create adaptive digital therapies. We close, discussing directions for future work.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Haptic devices**; **Touch screens**.

KEYWORDS

Digital Phenotype, Digital Therapy, ASD, Sensory Processing

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1 INTRODUCTION

Autism Spectrum Disorder (ASD) is characterized by persistent deficits in social and communication skills, as well as restrictive and repetitive behaviors [2]. In addition, 96% of children with ASD exhibit sensory difficulties [5], they can be hyper or hypo-sensitive, and 86% experience tactile sensitivity difficulties. For example, they do not like to be petted or carried, and they have an excessive fascination with certain textures and extreme avoidance of others [20]. Tactile processing is important to enable children to connect with those around them, feel sensations through their skin, and feel safe within such an environment [7, 13, 20].

A trained therapist can help improve sensory processing deficits using a therapy based on a sensory diet personalized to each child's needs. Such a sensory diet can increase alertness or arousal for optimal performance in daily tasks. A sensory diet is a tailored plan specifying the amount of time, duration, and intensity of a patient's exposure to one stimulus or a variety of stimuli, some are combined. The goals of the sensory diet are tailored individually [21] based on the appreciation of therapists and their interpretability based on experience and judgment. For example, tactile therapy can demand the daily exposure of a child to different touch sensations. When evaluating the sensory processing deficits of children with ASD, there are difficulties in finding appropriate measurement methods. Standardized tests are frequently unsuitable because children with ASD have diverse behavioral and developmental levels, making it challenging to maintain the validity of the tests. The available tests rely on observation, self-reports, and parental reports, however, there is no direct method of measurement in place. Moreover, when interventions like tactile stimulation treatments are customized to meet each child's unique requirements, it results in a wide range of goals and outcomes within a study. Therefore, it is vital to find measurement tools that accurately reflect the individualized needs and the significant variability observed in this population [19]. The field of Digital Therapeutics has proposed to use of wearables, mobile, and haptic interfaces to support tactile sensory stimulation. For example, FAR: End-to-End, is a distributed system that uses predefined or customized vibrotactile patterns to help children with ASD regulate their emotions through breathing [14]. A vibrotactile pattern is a function of intensity, rhythm, and sharpness representing the shape of a vibratory waveform. A wearable prototype

uses an inexpensive tactile sleeve with vibrotactile patterns to help individuals with ASD to learn how to manage hypersensitivity issues related to touch sensations. Such tactile stimulation can mimic everyday interactions, like a classmate's tap on your arm to get your attention or a comforting arm rub to show appreciation [24]. More recently, research has emphasized the capabilities of mobile technologies taking advantage of tactile stimulation received from touch interactions [3, 15, 16]. Altogether, this research has made the compelling argument that Digital Therapeutics, in particular mobile tactile stimulation, can gather continuous streams of data from touch and haptic interactions. However, little has been said if such data can be used as an objective and quantitative measurement to adapt and personalize Digital Therapeutic interventions. In this position paper, we present a proposal to use Digital Phenotypes as an approach to adjust the "digital active ingredients" from mobile tactile stimulation as key components to achieve efficacy. Our ongoing work focuses on developing Digital Phenotypes based on children's interaction with a haptic interface.

2 DEFINING DIGITAL PHENOTYPES FROM DIGITAL MARKERS

The field of Ubiquitous Computing has made significant progress to identify digital markers that quantitatively measure physiological or behavioral data collected through wearable devices, mobile sensors, and interactive surfaces [23]. These digital markers serve as the basis to develop a Digital Phenotype (DP) that represents continuous data collection from personal digital devices primarily from wearable or mobile devices, where the individual's interaction with digital technologies impacts every aspect of human health understanding, ranging from diagnosis and treatment to the management of chronic diseases [4, 10]. For example, researchers utilized a smart-phone app to gather touch interaction data over eight weeks, aiming to examine its correlation with sleep patterns [12]. An app recorded each tap on the screen, allowing estimation of daily phone usage and identification of potential periods of rest by detecting gaps in touch activity. Sleep episodes were defined as periods lasting 2-13 hours with minimal tap activity (less than 2 minutes of tapping within an hour). By analyzing this data, the researchers were able to identify five sleep digital markers: bedtime, wake time, mid-sleep time, time spent in bed, and sleep efficiency. These digital markers aided in characterizing distinct behavioral Digital Phenotypes associated with sleep and peri-sleep behaviors, enabling the development of personalized interventions targeting specific sleep-related issues. This study highlights how Digital Phenotypes of touch interactions can contribute to customized approaches for improving sleep behaviors. The field of Digital Phenotypes is relatively new, with most works focused on mental health and adult [17, 26], but little has been said about how to define the digital phenotype of ASD. Research on Digital Markers for ASD has been more active in identifying atypicalities in eye-gaze [6], voice [8], and motor coordination [1]; a recent trend has highlighted the importance of touch interactions to reveal digital markers of ASD [11]. These works have shown that spatial and temporal features of touch interactions [18], like frequency and duration of touch, can become a digital marker for ASD [1]. For example, Anzulewics found twenty digital markers related to the touch interactions (velocity, average gesture area,

standard deviation of height, maximum value of height, minimum duration of a touch gesture) and through the sensors being used. These digital markers contributed significantly to classifying the differences between children with ASD and neurotypical children [1].

On the other hand, the Bubble-Popping tablet-based game for the assessment of visual motor skills [18]. Touch gestures were analyzed from younger autistic versus neurotypical children, older autistic versus neurotypical children, and older autistic with and without ADHD. The authors identified 19 digital markers, such as number of touches, number of bubble pops, bubble popping rate, and touch frequency which showed potential for assessing early autism and measuring children's visual motor skills.

Table 1. A summary of most of the digital markers and their associated features in the literature. Our research has been

Digital Marker	Type	Feature	Description
Amplitude [11]	Spatial	$A = \sqrt{LAx^2 + LAy^2 + LAz^2}$	Amplitude vibration of the touch
Radius [11, 23]	Spatial	Touched radius	Touched radius
Force [11]	Spatial	$F = \text{Mass} \times \text{Acceleration}$	Force of the touch
Distance of Touch [22, 23]	Spatial	$D = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$	Length of the touch motion
Number of touches [23]	Temporal	$N = \text{Number of unique touches during task}$	Represents the total of unique touches on screen
Touch Duration [21-23]	Temporal	Δt	Time the finger is on the screen during a touch,
Velocity [10-12, 21-23]	Temporal	$V = \frac{\text{Distance of Touch}}{\text{Touch Duration}}$	Ratio between Distance of touch length and the Touch Duration,
Touch Frequency [23]	Temporal	$TF = \frac{N}{60 \text{ Seconds}}$	Number of touches on screen per second

focused on studying the capabilities of haptic interfaces to uncover digital markers of ASD, in particular, we showed that a haptic passive interface like an elastic surface can be used to identify digital markers of ASD related to touch interactions [3][16] and we are working on the design of a haptic mobile game that uses vibrotactile stimulation as a response to touch interactions [15]. The game specifically associates a vibrotactile pattern with touch gestures. The objective of the game is to help a little spider to build her web and feed the spider. A comparative analysis of the use of such a game by children with ASD and neurotypicals reveals several digital markers related to touch. The most relevant digital markers are related to the vibration intensity, the movement distance and direction, the tilt direction and velocity, the touch size and velocity and the touch space can be digital markers of ASD. Taking all together these digital markers can be combined to create a more robust representation of how an individual will more likely interact with touch-based interfaces and reveal further information on

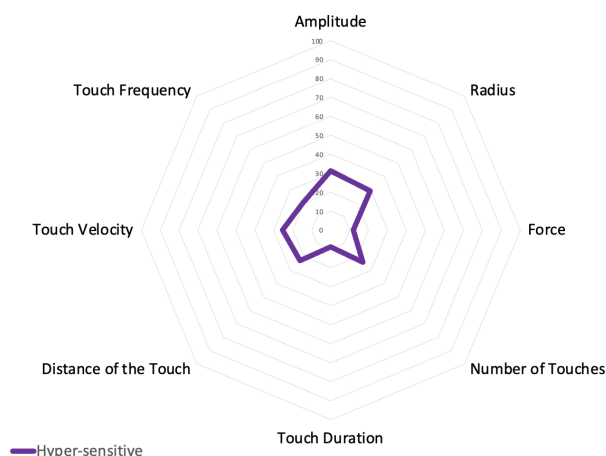


Figure 1: Illustrates a potential representation of the Digital Phenotype of hyper-sensitive children with ASD

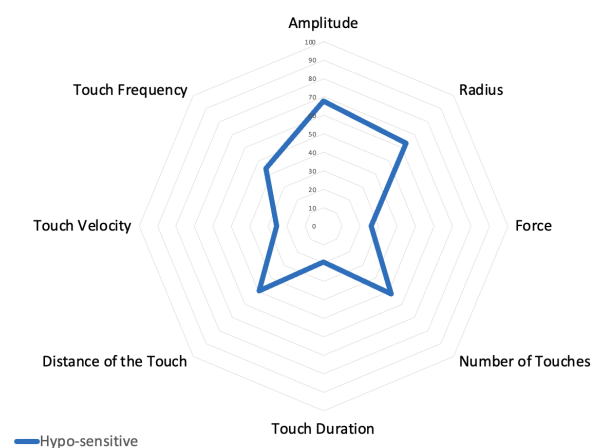


Figure 2: Illustrates a potential representation of the Digital Phenotype of Hypo-sensitive children with ASD

how such digital markers are connected to potentially become a Digital Phenotype (Figures 1,2,3). These bodies of work serve as the foundation for further studies aimed at developing objective tools to understand the diverse development trajectories of children with ASD. Our work will focus on two objectives to address tactile sensory processing difficulties in ASD preschoolers. The first is to develop a Digital Phenotype patient trajectory, while the second objective is to utilize the established Digital Phenotype in adaptive therapy.

3 DIGITAL PHENOTYPE FOR PATIENT TRAJECTORY

For therapists and parents of children with ASD, it is important to understand their children's starting point and track their progress

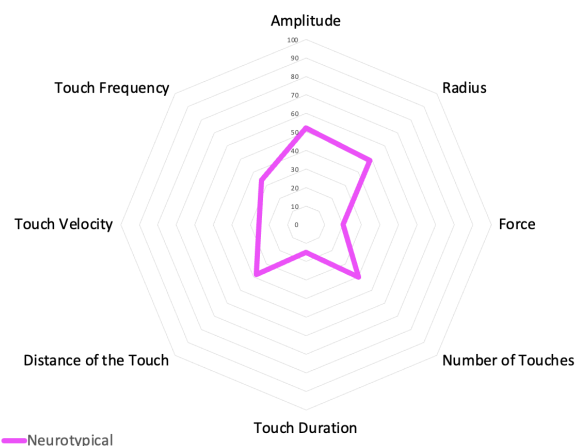


Figure 3: Illustrates a potential representation of the Digital Phenotype for neurotypical children

as they do the activities during the sensory tactile therapy sessions [21]. Traditionally, progress measurements present inherent difficulties in identifying appropriate measurement methods for tactile sensory processing deficits. Standardized measures frequently prove inadequate due to the heterogeneous behavioral and developmental levels observed among children with ASD, thereby impeding the maintenance of test validity [19]. Thus, the opportunity of using haptic interfaces to objectively monitor the behavior and track patient trajectory progress in the therapy. To address this concern, we propose to develop a Digital Phenotype based on analyzing the 42 digital markers identified in a smart tablet [1], haptic mobile game [15], and Bubble-Popping [18]. Once we have identified the significant digital markers that are needed to establish the Digital Phenotype of tactile processing, we will collect data with the haptic mobile game [15] to further study the data to obtain appropriate boundaries that allow us to understand the range and number of different levels of tactile sensory processing needs from hypo or hyper-sensory processing needs to the opposite neurotypical developed.

For example, various individuals who are "Hypersensitive" based on the markers from Table 1 could be represented as it is illustrated in Figure 1, with Vibration (Amplitude), Radius (touch size), force, number of touches, touch duration, distance of the touch, touch velocity and touch frequency. We hypothesize that a hypersensitive child would not have low amplitude, a small irregular radius, and little force. In contrast, a hypo-sensitive child would have the opposite values as depicted in Figure 2, all within the limits that are for children with ASD. In the future when a child will attend a therapy session, instead of the traditional observations or parental reports, this research will allow therapists to quantitatively measure tactile sensory processing. For instance, during the session the child will engage in the haptic game, with the collected data and the therapist will observe the child's Digital Phenotype similar to Figure 1. and will tailor a sensory diet, determining the optimal duration, intensity, and frequency of exposure to the stimuli. This approach will enhance therapeutic intervention to guide the progress of the

patient trajectory to modify their Digital Phenotype very similar to neurotypical children as it is illustrated in Figure 3.

4 DIGITAL PHENOTYPE FOR ADAPTIVE THERAPY

Due to the diversity of the tactile processing issues each child has a particular need and therapy should be adapted. Our approach will be on children's tactile sensory processing needs through haptic interfaces which are categorized as passive or active, with the latter being capable of stimulating the human tactile sensory system with forces, ultrasonic waves, and vibration. The vibrations can generate different sensations on the skin through what are known as vibrotactile patterns. Research has focused on designing vibrotactile pattern libraries [9, 22, 25]. One example is VibBiz, an interactive web tool library comprising 120 vibrotactile patterns classified into five categories: physical, sensorial, emotional, metaphor, and use. For this purpose, the categories of sensorial and emotional vibrotactile patterns are particularly relevant, as they can be linked to the hyper- or hyposensitivity experienced by individuals with ASD. Therefore, it is crucial to study the tolerance levels for specific vibrotactile patterns that can be classified as smooth or rough in this context. To address this, our approach involves combining vibrotactile patterns and the Digital Phenotype to create a new haptic game. This game will be specifically designed for digital tactile therapy, allowing for personalized treatment. This includes determining the "digital active ingredients" of the digital therapy as the frequency of the therapy sessions, such as daily or twice a day, setting the duration of each session, and establishing the number of weeks for the overall treatment. Our aim is to tailor the therapy to individual needs. In our vision, children will actively engage in the haptic game and enjoy various tactile activities. Throughout the game, we will collect interaction data, enabling us to continuously assess their progress. The digital phenotype approach allows us to automatically adapt the digital therapy based on response to the patient trajectory and personalize the intensity function, rhythm, and sharpness of the vibrotactile pattern. With this approach children will enhance their sensory perceptions and abilities, ensuring optimal outcomes moving forward from hypo, hypersensitive Digital Phenotypes as Figure 1-2 to neurotypical development as Figure 3.

5 CONCLUSION AND FURTHER WORK

The use of Digital Phenotypes, touch-based and haptic interfaces present a promising avenue for measuring and improving tactile sensory processing deficits in children with ASD. This approach has the potential to enhance the objectivity, accuracy, and personalization of Digital therapeutic interventions, ultimately improving the quality of life for individuals with ASD and their families. Although there have been several digital markers proposed in the literature (Table 1), it is crucial to acknowledge that the current understanding, especially that of related to our own research, is in its initial stages and we need to gather more data. Furthermore, we aim to evaluate the relationships between such digital markers and to what extent they could be used to characterize Digital Phenotypes. In this position paper, we argue that if uncovered, such Digital Phenotypes can be effective to follow up on patient trajectories and to create adaptive digital interventions. In future work, we will

explore tolerance levels for vibrotactile patterns that can be categorized as either smooth or rough. This investigation will help us understand how these patterns relate to the tactile characteristics of individuals with ASD. Lastly, we will facilitate design sessions involving expert therapists. Their valuable insights will guide us in developing the haptic game for tactile digital therapy, ensuring that it aligns with the specific needs and preferences of individuals with ASD.

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